VERDANT INTELLIGENCE

Agricultural Plant Disease Outbreak Forecasting & Surveillance System for India

Phase 2 Mid-Semester Comprehensive Evaluation Report & Faculty Oral Defense Guide

Course / Subject:	Time Series Analysis & Forecasting (SEM-5)	**Evaluation Window:**	October 19 – 30, 2026
Evaluation Phase:	Phase 2: Mid-Semester Evaluation (10 Marks)	**Verification Status:**	**Fully Implemented & Verified** (21/21 Tests Passed)
Core Pathogens:	10 Crop Pathogens (GBIF 2015–2026, 3,076 records)	**Meteorology:**	NASA POWER (132 continuous months joined)

1. Executive Summary & Evaluation Rubric Alignment

Verdant Intelligence is an econometric and machine-learning intelligence platform designed to model, test for stationarity, and forecast monthly agricultural plant disease report trajectories across India. The project addresses fungal and bacterial pathogens threatening staple crops (Rice, Groundnut, Pigeonpea, Chickpea, Wheat, Mustard) by combining real-world Global Biodiversity Information Facility (GBIF) spatial occurrences with NASA POWER meteorological reanalysis. The system has been developed, tested, and demonstrated end-to-end to fulfill every requirement of the Phase 2 Mid-Semester Rubric:

Evaluation Rubric Component	Project Implementation Details & Technical Output	Module / File	Marks
1. Classical Time Series Models	ARIMA(1,1,1), SARIMA(1,0,1)(1,0,1)[12], Holt-Winters Triple Additive, Seasonal Naive Baseline, AR(1), MA(1).	Page 3 & 4 src/forecasting.py	**3 Marks**
2. Data Preprocessing & Stationarity	Continuous monthly resampling (`MS`), zero vs. missing distinction, joint ADF and KPSS tests, first differencing (d=1).	Page 2 src/stationarity.py	**3 Marks**
3. Model Comparison & Metrics	11-model out-of-sample benchmark (RMSE, MAE, zero-safe sMAPE, WAPE). Parsimony proof: ARIMA(1,1,1) beats Seasonal Naive & ML.	Page 3 src/evaluation.py	**3 Marks**
4. Observations & Interim Roadmap	Biological t-1 weather incubation lag interpretation, Kharif/Rabi dynamics, Phase 3 spatial downscaling to district level.	Page 2, 4 & 5 Evaluation Report	**1 Mark**

2. Data Provenance & Cleaning Pipeline

- **Strict Data Provenance:** Historical occurrences were curated from GBIF covering 10 major Indian crop pathogens (2015–2026), geofenced within India (6.0°N ≤ Lat ≤ 37.5°N, 68.0°E ≤ Lon ≤ 97.5°E). Weather parameters (T2M, TMAX, TMIN, PRECTOTCORR, RH2M) were ingested from NASA POWER at continuous monthly frequency (132 continuous months).
- **ISO 8601 Temporal Parsing:** Validated collection event dates versus database deposit dates; eliminated duplicate records and rejected future timestamps.
- **Zero vs. Missing Epistemological Distinction:** In observational biodiversity surveillance, an unsurveyed month does NOT equal genuine zero infections. During monthly resampling (`MS`), missing intervals are flagged (`is_missing=True`) rather than artificially padded with zeros, preventing severe downward bias in moving averages.
- **Outlier Z-Score Inspection:** Outbreak spikes are verified against historical agrometeorological events (e.g., monsoon surges) rather than trimmed as sensor noise.

3. Econometric Stationarity Testing & Parameter Identification

Weak Stationarity Prerequisite: Classical autoregressive models require constant mean, constant variance, and autocovariance independent of time. If a series exhibits a unit root, random environmental shocks cause variance to explode over time. We implemented a **joint stationarity framework** combining the Augmented Dickey-Fuller (ADF) test (H_0: unit root present) and the KPSS test (H_0: trend-stationary):

Pathogen / Crop Host	Raw ADF Stat (p-value)	Raw KPSS Stat (p-value)	Joint Decision	Transformation	Differenced ADF (p-value)
Macrophomina phaseolina (Dry Root Rot)	+0.0915 (p = 0.9655)	1.1553 (p = 0.0100)	**Non-Stationary (Unit Root)**	First Diff (d = 1)	**-9.4821 (p < 0.0001)**
Pyricularia oryzae (Rice Blast)	-1.8420 (p = 0.3598)	0.8421 (p = 0.0100)	**Non-Stationary (Unit Root)**	First Diff (d = 1)	**-8.7214 (p < 0.0001)**
Fusarium oxysporum (Wilt Complex)	-2.1104 (p = 0.2403)	0.9812 (p = 0.0100)	**Non-Stationary (Unit Root)**	First Diff (d = 1)	**-7.8940 (p < 0.0001)**

Stationarity Proof: All raw series fail the ADF test ($p > 0.05$) and reject the KPSS null ($p < 0.05$). Applying first-order differencing ($y'_t = y_t - y_{t-1}$) drops the ADF p-value to < 0.0001, formally proving that first-order integration (**d = 1**) eliminates stochastic drift.

4. Classical Forecasting Model Formulations & Assumptions

- **Seasonal Naive Baseline:** $y_t^{pred} = y_{t-12}$. Predicts current incidence based entirely on the corresponding month of the prior annual cropping cycle. Serves as the mandatory benchmark to prove model skill over simple annual repetition.
- **ARIMA(1, 1, 1):** $(1 - \phi_1 B)(1 - B)y_t = (1 + \theta_1 B)\varepsilon_t$, where $\varepsilon_t \sim WN(0, \sigma^2)$. Captures momentum via AR(1) and moving-average shock absorption via MA(1) on differenced stationary data. Selected via ACF/PACF cutoffs and verified through Ljung-Box white noise testing ($p > 0.05$).
- **SARIMA(1, 0, 1)(1, 0, 1)[12]:** Incorporates multiplicative seasonal lag polynomials with period s = 12 months, capturing annual Kharif (monsoon) and Rabi (winter) epidemiological cycles.
- **Holt-Winters Triple Exponential Smoothing (Additive):** Decomposes the series into Level (l_t), Trend (b_t), and Seasonality (s_t) using recursive smoothing parameters $\alpha, \beta, \gamma \in [0, 1]$. Adapts dynamically without rigid Gaussian disturbance assumptions.
- **ARIMAX / SARIMAX with Lagged Weather:** $y_t = \sum \beta_k X_{k, t-1} + [\Theta(B)/\Phi(B)]\varepsilon_t$. Uses 1-month lagged agrometeorology (X_{t-1}: rainfall, temperature, relative humidity). Avoids future data leakage while modeling the biological incubation lag.
- **Machine Learning Autoregressors:** HistGradientBoostingRegressor and RandomForestRegressor trained on autoregressive lag features (t-1, t-2, t-3, t-12).

5. Chronological Out-of-Sample Benchmark & Model Comparison

Zero Lookahead Data Leakage Protocol: Models were trained strictly on the initial 70% chronological timeline (2015–2022) and evaluated against the held-out 30% horizon (2023–2026). No random cross-validation shuffling was permitted. 11 distinct model architectures were evaluated on identical test intervals:

Rank	Forecasting Model Architecture	RMSE	MAE	MAPE (%)	sMAPE (%)	WAPE (%)	Bias
#1	**ARIMA(1, 1, 1)**	**1.823**	**1.243**	68.4%	123.45%	**85.73%**	+0.34
#2	Seasonal Naive Baseline ($y_t = y_{t-12}$)	1.871	1.400	82.1%	129.14%	96.55%	-0.15
#3	ARIMAX (Lagged Weather t-1)	1.873	1.298	71.2%	133.37%	89.51%	+0.38
4	SARIMAX (Seasonal + Lagged Weather)	1.873	1.290	70.8%	131.89%	88.99%	+0.22
5	Holt-Winters Triple Exponential Smoothing	1.889	1.301	74.5%	119.57%	89.73%	+0.42
6	Hist Gradient Boosting Regressor	1.959	1.383	79.1%	151.64%	95.38%	+0.51
7	SARIMA(1,0,1)(1,0,1)[12]	2.012	1.346	76.3%	153.34%	92.83%	+0.48
8	AR(1) Autoregressive	2.201	1.457	84.6%	161.78%	100.48%	+0.58
9	MA(1) Moving Average	2.202	1.455	83.9%	140.00%	100.34%	-0.62
10	Random Forest Autoregressor (Lags 1,2,3,12)	2.211	1.661	91.2%	166.35%	114.55%	+0.65
11	Meta Prophet (Additive Fourier Seasonality)	2.427	1.596	88.5%	133.89%	110.07%	+0.72

6. Statistical Parsimony & Metric Integrity Policy

- **Statistical Parsimony Proof (Occam's Razor):** ARIMA(1,1,1) outperforms complex machine learning models (Gradient Boosting, Random Forest) because monthly epidemiological counts contain low counts and intermittent zeros. Tree models overfit idiosyncratic local noise and fail to extrapolate beyond historical splits. ARIMA(1,1,1) estimates only 3 parameters (ϕ_1, d=1, θ_1), delivering the lowest RMSE (1.823) and MAE (1.243).
- **Zero-Denominator Policy (Why Raw MAPE Fails):** In disease surveillance, actual counts y_t can be zero. Raw MAPE involves division by zero (1/0 = ∞). Silently adding arbitrary constants produces misleading percentages. Verdant Intelligence computes:
  1. **Symmetric MAPE (sMAPE ≤ 200%):** Denominator $(|y_t| + |y_t^{pred}|)/2$ bounds percentage error safely.
  2. **Weighted Absolute Percentage Error (WAPE):** $\Sigma |y_t - y_t^{pred}| / \Sigma y_t$. ARIMA(1,1,1) achieves WAPE = 85.73%, beating the Seasonal Naive baseline (96.55%) by 10.8 percentage points.
- **Non-Negative Forecast Enforcement:** Point forecasts and confidence intervals are bounded at zero via $\max(0, y_t^{pred})$ in src/forecasting.py, preventing impossible negative disease report counts.

7. Plant Pathology & Agronomic Domain Interpretation

A forecasting model in agriculture is only valuable if its outputs reflect true biological mechanisms. Verdant Intelligence grounds all quantitative outputs in Indian agro-ecological realities:

- **The Biological Incubation Lag (t-1 Peak Cross-Correlation):** Cross-correlation analysis between NASA POWER precipitation (PRECTOTCORR) and Rice Blast (*Pyricularia oryzae*) reports reveals a statistically significant peak at **Lag t-1 (r = +0.58, p < 0.01)**. In plant pathology, fungal conidia require >10 hours of continuous leaf wetness and relative humidity >85% for germination and appressorium penetration. The 1-month lag reflects the latent incubation period required for fungal mycelium to colonize host tissues and manifest macroscopic necrotic lesions observed during field surveillance.
- **Thermal Stress & Drought Dynamics (Dry Root Rot):** For *Macrophomina phaseolina*, incidence accelerates when soil temperatures exceed 32°C combined with moisture deficit (May–June pre-monsoon and post-monsoon dry intervals). The model accurately detects these thermal stress vulnerability windows.
- **Operational Extension Preemption Window:** Generating 3-to-12 month forward projections with 80% and 95% confidence intervals gives Krishi Vigyan Kendras (KVKs) and ICAR a 30-to-60 day operational window to preposition systemic fungicides and broadcast preventative advisories.
- **Scientific Data Caveat:** All dashboard displays explicitly clarify that figures represent *recorded scientific/field occurrences*, not total infested crop acreage. This preserves scientific accuracy and prevents sensationalist misinterpretations.

8. Software Engineering & 6-Module Streamlit Architecture

The platform is organized into an enterprise-grade modular Python package running live on port 8501:

Module / Page	Primary File Path	Key Features & Scientific Capabilities
1. Overview	app.py	Executive hero section, high-level dataset metrics (3,076 records, 10 pathogens), best-model preview (ARIMA(1,1,1)), 6-page navigation.
2. Disease Explorer	pages/1_Disease_Explorer.py	Pathogen selector, host crops, geographical distribution map, monthly timeline, 3-part interpretation cards.
3. Time Series Analysis	pages/2_Time_Series_Analysis.py	8 comprehensive analytical tabs: Resampling, Outlier Z-score inspection, Seasonality heatmaps, STL decomposition, ACF/PACF memory, Dual Stationarity (ADF+KPSS), NASA POWER Weather Lags (t-1), Chronological split.
4. Model Comparison	pages/3_Model_Comparison.py	11-model benchmark table, RMSE/MAE/sMAPE bar charts, Actual vs Predicted overlays, Error distribution plots, Ljung-Box white noise diagnostics, Mathematical parameter assumptions tab.
5. Future Forecasts	pages/4_Future_Forecasts.py	3-12 month forward projections, 80% & 95% confidence intervals, non-negative bounds (zero clipping), Section 9 FORECAST SUMMARY card answering all 5 operational questions.
6. Data & Methodology	pages/5_Data_and_Methodology.py	GBIF & NASA POWER data documentation, ISO 8601 parsing rules, stationarity methodology, limitations, data download center.

Automated Verification Suite: 21 automated unit tests in tests/test_all_features.py pass in 0.64s, validating data pipelines, statistical stationarity modules, model convergence, and zero-safe metric calculations.

PART 2: FACULTY ORAL VIVA & EXAMINATION DEFENSE GUIDE

Authoritative High-Scoring Responses to Anticipated Evaluation Panel Questions

Curated for the Phase 2 Mid-Semester Evaluation across Time Series Econometrics, Modeling, and Data Integrity.

Theme A: Time Series Stationarity, Transformations & Diagnostics

Q1. Why is stationarity testing mandatory before fitting ARIMA models? What does a 'unit root' mean physically in an epidemic?

Classical autoregressive models assume weak stationarity: constant mean, constant variance, and autocovariance depending only on time displacement (h). If a time series contains a unit root ($|z| = 1$), any random shock—such as an unseasonal cyclone or disease super-spreader event—persists indefinitely, causing variance to explode as $t \to \infty$. Physically in plant pathology, a non-stationary epidemic means disease momentum compounds cumulatively across seasons. First differencing ($d = 1$) removes this stochastic integration, transforming the cumulative level into month-over-month infection velocity, which has stable variance.

Q2. Why did you implement BOTH the Augmented Dickey-Fuller (ADF) test AND the KPSS test?

ADF and KPSS have complementary null hypotheses, protecting us from false conclusions: ADF tests H_0: 'The series has a unit root (non-stationary)' vs. H_1: 'Stationary'. KPSS tests H_0: 'The series is level/trend stationary' vs. H_1: 'Unit root present'. If ADF fails to reject ($p > 0.05$) AND KPSS rejects ($p < 0.05$), we have conclusive mathematical proof of a stochastic unit root. In our data, raw *Macrophomina* showed ADF $p = 0.9655$ and KPSS $p = 0.0100$. After first differencing ($d = 1$), ADF $p < 0.0001$, confirming strict stationarity.

Q3. How did you determine the parameter orders (p, d, q) for ARIMA(1,1,1)?

We used a 3-step identification protocol: 1. Integrated order ($d = 1$) was determined by the joint ADF-KPSS test. 2. Order selection (p, q) was guided by sample ACF and PACF correlograms: PACF showed a sharp cutoff after lag 1 (indicating $p = 1$), while ACF exhibited significant correlation at lag 1 with exponential damping (indicating $q = 1$). 3. We systematically compared AIC and BIC criteria across grid orders (0 to 3). ARIMA(1,1,1) yielded the lowest BIC while passing the Ljung-Box test for white-noise residuals ($p > 0.05$).

Q4. How do you distinguish between an unsurveyed month and an active zero-infection month?

In observational biodiversity data like GBIF, observers record positive identifications. If month M has zero records, it may represent true absence (zero disease) OR an absence of sampling effort (e.g., lockdown, monsoon impassability). When resampling to monthly start frequency (`MS`), we preserve survey metadata (`is_missing=True`). For econometric estimation, we do not impute arbitrary zeros without justification, as doing so would artificially deflate the moving average baseline and introduce severe under-forecasting bias.

Theme B: Econometric vs. Machine Learning Forecasters & Model Assumptions

Q5. Why did classical ARIMA(1,1,1) outperform modern machine learning models like Random Forest and Gradient Boosting?

This demonstrates the principle of **Statistical Parsimony (Occam's Razor)**. Machine learning regressors (Random Forest, Gradient Boosting) rely on non-parametric decision trees that require large sample sizes (thousands of observations) to partition feature space effectively. In monthly macro-epidemiology (132 time points), tree models overfit idiosyncratic noise and cannot extrapolate trend outside their historical training leaf values. ARIMA(1,1,1) estimates only 3 parameters (ϕ_1, $d=1$, θ_1), giving it superior generalization on sparse chronological holdout data (RMSE 1.823 vs. Random Forest 2.211).

Q6. Why is the Seasonal Naive model your baseline, and did your models beat it?

In agricultural time series with annual cropping seasons (Kharif and Rabi), a naive random walk ($y_t = y_{t-1}$) is inappropriate because winter weather does not resemble monsoon weather. The correct, rigorous benchmark is the Seasonal Naive model ($y_t = y_{t-12}$), which predicts that this year's July disease incidence will equal last year's July incidence. Our ARIMA(1,1,1) successfully defeated the Seasonal Naive baseline, achieving an RMSE of 1.823 vs 1.871, MAE of 1.243 vs 1.400, and a 10.8 percentage point improvement in WAPE (85.73% vs 96.55%).

Q7. What are the key mathematical components of Holt-Winters Exponential Smoothing?

Holt-Winters Triple Exponential Smoothing decomposes the series into Level (l_t), Trend (b_t), and Seasonality (s_t) using smoothing coefficients $\alpha, \beta, \gamma \in [0, 1]$:

- Level: $l_t = \alpha(y_t - s_{t-m}) + (1 - \alpha)(l_{t-1} + b_{t-1})$
- Trend: $b_t = \beta(l_t - l_{t-1}) + (1 - \beta)b_{t-1}$
- Season: $s_t = \gamma(y_t - l_t) + (1 - \gamma)s_{t-m}$

Its strength is adaptive exponential decay weighting without requiring strict Gaussian disturbance assumptions. It ranked 5th overall (RMSE 1.889) and achieved the lowest sMAPE (119.57%).

Q8. How does ARIMAX/SARIMAX incorporate weather variables without creating data leakage?

In ARIMAX, exogenous variables X_t must be known at forecast time t. If we used contemporaneous weather (X_t), we would be predicting future disease using unknown future weather, which constitutes temporal data leakage. We solved this by using **lagged exogenous variables (X_{t-1})**: last month's precipitation, temperature, and humidity. This is both mathematically valid (known at time t) and biologically sound, as fungal spore germination takes 2–4 weeks to develop into detectable field lesions.

Theme C: Metric Rigor & Sparse Data Mathematical Integrity

Q9. Why is standard Mean Absolute Percentage Error (MAPE) mathematically problematic in this dataset?

The formula for MAPE is $(1/n) \Sigma |(y_t - y_t^{pred}) / y_t| \times 100\%$. When actual occurrence count $y_t = 0$, the denominator is zero, making MAPE mathematically undefined (∞). Silently replacing zero with an arbitrary epsilon like 0.001 produces wildly inflated, distorted percentages (e.g., 50,000%). To uphold scientific integrity, Verdant Intelligence computes:

1. **sMAPE (Symmetric MAPE):** Bounds percentage error between 0% and 200% via denominator $(|y_t| + |y_t^{pred}|)/2$.
2. **WAPE (Weighted Absolute Percentage Error):** $\Sigma |y_t - y_t^{pred}| / \Sigma y_t$, which aggregates deviations over total incidence, eliminating zero-division entirely.

Q10. Why do you report both RMSE and MAE? What does a large gap between them indicate?

MAE evaluates linear average error (L_1 norm), weighting all deviations equally. RMSE squares errors before averaging (L_2 norm), heavily penalizing large forecast misses. In epidemiological surveillance, failing to predict a massive disease outbreak is far more damaging to farmers than a small 1-unit baseline error. The fact that ARIMA(1,1,1) achieved the lowest RMSE (1.823) confirms it does not produce catastrophic outlier spikes during non-epidemic months.

Theme D: Agricultural Domain Interpretation & Plant Pathology

Q11. What does the 1-month weather lag (t-1) tell us about Rice Blast biology?

Rice Blast is caused by the hemibiotrophic fungus *Pyricularia oryzae*. Conidia germinate on leaf surfaces only when relative humidity exceeds 85–90% and free moisture (dew/rain) persists for at least 10–14 hours at 20–28°C. After appressorium penetration, the fungal mycelium undergoes an asymptomatic biotrophic phase inside plant tissue before transitioning to necrotrophic lesion expansion. This latent incubation period takes approximately 2 to 4 weeks. Therefore, high rainfall in July drives macroscopic foliar necrosis and field observations recorded in August (Lag t-1).

Q12. How does this system differentiate between 'recorded disease occurrences' and 'infected crop acreage'?

This is a fundamental scientific caveat in our platform. Biodiversity and herbarium databases (GBIF) record confirmed research accessions and field observations. They measure *disease presence and spatial reporting intensity*, NOT the percentage of farm acreage destroyed. Our user interface and documentation explicitly state: *'Values represent recorded research occurrences. They serve as an early-warning outbreak proxy, not total destroyed agricultural acreage.'*

Theme E: Software Verification & Phase 3 Roadmap

Q13. How did you verify code correctness and prevent regressions?

We built an automated test suite using `pytest` located in `tests/test_all_features.py`. The suite executes 21 automated tests verifying: data loading pipelines, date conversions, continuous monthly aggregation, missing value imputation, ADF/KPSS statistical testing, ARIMA/SARIMA/Holt-Winters execution, error metric calculations (verifying zero-safe MAPE/sMAPE/WAPE), and UI helper rendering. All 21 unit tests pass in 0.64 seconds with zero syntax errors across the entire codebase.

Q14. What is planned for Phase 3 to build upon these Phase 2 achievements?

Phase 2 successfully established the econometric baseline, stationarity proofs, classical benchmark, and weather lag mechanisms at macro-regional scale. For Phase 3 (End-Semester), we will implement:

1. **District-Level Spatial Downscaling:** Geographically localized hierarchical forecasting downscaled to top agricultural districts.
2. **Deep Sequence Architectures:** Benchmarking Temporal Fusion Transformers (TFT) and N-BEATS with multi-pathogen cross-learning.
3. **Automated Early-Warning Push API:** Automated webhook dispatching real-time risk alerts to farmer cooperatives.

EXECUTIVE 5-MINUTE ORAL VIVA PITCH (MEMORIZE FOR EVALUATION OPENING):
'Respected Evaluators, Verdant Intelligence addresses the critical challenge of forecasting crop pathogen outbreaks across India. Our empirical pipeline combines 11 years of GBIF disease records with NASA POWER agrometeorology across 132 continuous months. Through joint ADF and KPSS tests, we proved raw pathogen series exhibit stochastic unit roots, establishing that first differencing (d=1) is mathematically essential. Across an 11-model out-of-sample benchmark evaluated on held-out data with zero lookahead leakage, classical ARIMA(1,1,1) achieved top performance (RMSE 1.823, MAE 1.243), outperforming the Seasonal Naive baseline by 10.8 percentage points in WAPE. Crucially, we demonstrated that simple parsimonious econometrics outperform complex decision trees on sparse count data, and cross-correlation confirmed a 1-month biological incubation lag between monsoon precipitation and macroscopic foliar blast lesions. The system is fully operational across 6 Streamlit modules and validated by 21 passing automated unit tests.'